KEVIN TA

MECHANICAL ENGINEERING

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Education

ETH Zürich
M.Sc in Robotics, Systems, and Control
Incoming September 2020

University of British Columbia
B.ASc in Mechanical Engineering
Mechatronics Specialization
Graduate of 2020
Major GPA: 4.0 (91%)

Technical Skills

Mechanical
SolidWorks (Certified Associate) · Simulation Tools · Finite Element Analysis · Mechanical Design · Composite Materials · 3D Printing · Machine Tools

Software
Python · C/C++ · MATLAB · SQL· LaTeX · GIS · Tableau

Electrical
Arduino · Raspberry Pi · Sensors & Actuators · Electronic Device Specification · Function Generator · Oscilloscope · Soldering · Altera FPGA

Research

UBC CARIS (Robotics) Laboratory — May 2019 - Present

MECHATRONICS RESEARCH ASSISTANT

- Developed wireless sensor data acquisition systems for a wheelchair, including collection of inertial measurement unit (IMU) to reliably sample data at 300 Hz through Bluetooth and Wi-Fi
- Utilized a combination of Python and C++ for data acquisition software to establish data transfer integrity and reconstructed time synchronization across various sensors
- Initiated experimental work in user motion intention, terrain classification, and terrain slope based on the environmental impulse excitations gathered by the data acquisition systems
- Researched mechanical phenomena such as caster flutter and rigid body kinematics for rationalizing terrain machine learning classifications

Centre for Hip Health and Mobility — May – Aug. 2016

UNDERGRADUATE RESEARCH ASSISTANT

- Collected dynamic data with accelerometers in high impulse linear impact testing to determine optimal helmet padding material and configurations for preventing head trauma
- Investigated statistical trends for x-ray machinery to determine long-term consistency for bone mass density studies
- Operated and maintained medical imaging tools based on fluoroscopy including DXA, CT, and C-arm machines for internal testing of radiation protection equipment and bone phantom samples
- Ran data collection sessions and interviews with elderly at-risk populations to assess how group activity encourages healthier and more mobile lifestyles

Teaching

UBC Dept. of Mechanical Engineering — Jan. – Apr. 2019

UNDERGRAUDATE TEACHING ASSISTANT

- Instructed, evaluated, and provided feedback to second year students through thermodynamic and fluid dynamic lab experiments to demonstrate core mechanical engineering topics
- Performed lab lectures to go over key concepts explored in lab experiments, ensuring students understood which mechanical phenomena to note in report analysis
- Developed lab process and data collection procedures to ensure students appropriately observed and analyzed

Awards & Achievements

UBC MECH Leadership Award 2020

Awarded by the UBC Mechanical Engineering Department for maintaining an A-grade average

UBC MECH Academic Achievement Award 2016 – 2019

Awarded: 4x

Awarded by the UBC Mechanical Engineering Department for maintaining an A-grade average

Trek Excellence Scholarship 2015 - 2018

Awarded: 4x

Awarded by the University of British Columbia for finishing in the top 5% of students

Dean's Honour List 2015 - 2018

Awarded: 5x

Awarded for exceptional academic achievement in the faculty of Applied Science

Donald J. Evans Scholarship in Engineering 2018

Awarded by Faculty of Applied Science based on academic merit

University of British Columbia Scholarship 2018

Awarded by the University of British Columbia based on academic achievement

S. Cyril Maplethorp Memorial Scholarship in Engineering 2017

Awarded by Faculty of Applied Science based on academic merit

Talisman Energy Scholarship 2016

Awarded for proven commitment to the energy sector and outstanding academic achievement

APEGBC Post-Secondary Entrance Scholarship 2014

Awarded by the Association of Professional Engineers and Geoscientists for outstanding achievement

British Columbia Government Scholarship 2014

Awarded by the government of BC for achieving exam results in the top 20 for the province of British Columbia

Industry Work Experience

Schneider Electric Solar — Jan. – Aug. 2018

SOLAR PREDICTIVE ANALYTICS AND MODELLING INTERN

- Implemented machine learning anomaly detection algorithms to analyze incoming data from on-site inverters for predictive and preventative maintenance reliability models
- Developed algorithms in Python to more accurately capture annual temperature profiles by querying 30 arc-second GIS weather data for geographical idiosyncrasies
- Validated thermal modelling, predictive algorithms, and service predictions for Schneider Electric Solar Inverters to ensure internal and external reliability data aligned with cloud data from on-site inverters
- Conducted stress tests and automated data collection of thermal and electrical characteristics of solar utility inverters

Pacey MedTech — May – Aug. 2017

MECHANICAL ENGINEERING INTERN

- Designed a novel urinary continence device through rapid prototyping, clinical trials, and low-quantity production to meet accelerated 3-month product development deadline
- Created and maintained documentation and manufacturing drawings to comply with Health Canada and FDA regulation using ISO 9001
- Designed injection-molded parts for manufacturing with consideration to parting lines, drafting, and symmetry

Smith + Andersen — Sep. – Dec. 2016

JUNIOR MECHANICAL DESIGNER

- Specced electrical and mechanical HVAC equipment requirements to suppliers and on-site contractors to optimize thermal performance and building systems efficiency
- Calculated heating, cooling, and ventilation loads based on building location, room usage, and building design

UBC Supermileage

Captain — Jun. 2019 - Present

- Acquired and managed an operational budget of over $100,000 through university and external organization funding applications to support diverse engineering projects
- Led 65 students comprising the UBC Supermileage Team to further improve vehicle efficiency through power transmission performance and drag resistance reduction to achieve 3000 MPG for our prototype vehicle
- Researched and implemented structural and dynamic systems that comply with strict engineering guidelines for two international competitions
- Encourage the diversification and development of new clean energy technology in the pursuit of the Electric Urban Concept and the Hydrogen Fuel Cell Prototype vehicles to inspire a generation of students focused on the challenges associated with climate change and green technology

Vehicle Mechanics Division Lead	Jun. 2018 – Jun. 2019
Safety Officer	Sep. 2017 – Aug. 2018
Aerodynamics Division Lead	Sep. 2016 – Jun. 2018
Aerodynamics Division Member	Sep. 2015 – Aug. 2016

UBC Supermileage Competition

Shell Eco-Marathon Americas

2019	2nd Place	Gasoline Prototype	1371 MPG
2019	-	Electric Urban Concept	Passed Technical Inspection
2018	6th Place	Gasoline Prototype	960 MPG
2018	-	Gasoline Urban Concept	Passed Technical Inspection

SAE Supermileage

2020	4th Place	Gasoline Prototype	Virtual Competition
2019	2nd Place	Gasoline Prototype	2229 MPG
2018	3rd Place	Gasoline Prototype	1405 MPG
2018	Design Excellence Award	Gasoline Prototype	

Additional Experience

Attentiv Medical — Sep. 2019 – Jun. 2020

FOUNDER & MECHATRONICS ENGINEER

- Validated medical problem for IV infiltration in neonates as a viable entrepreneurial start-up in collaboration with three cross-disciplinary engineers and two business students
- Developed market strategy, market potential, and a 10-year business plan to develop the idea from conception to acquisition
- Performed economical proof-of-concept testing with simulated models in pork and blood as well as gelatin and saline fluid, culminating in salvage pig cadaver tests
- Achieved high accolades in various start-up competitions and programs for depth of validation and early prototyping

Penguin Robotics — Sep. 2016 – Jan. 2017

VOLUNTEER

- Developed curriculum for robotics and STEM outreach to inspire and encourage underserved youth in the Lower Mainland to consider careers in STEM
- Acted as liaison for corporations and organizations wishing to contribute to Penguin Robotics' STEM outreach
- Aided setup at competition venues and performed outreach to secondary school students for the Lower Mainland Vex Robotics Circuit to promote STEM and robotics to youth